Indirect Self-tuning Regulators (Åström & Wittenmark, Examples 3.4–3.6)

Adaptive Control, Homework 3

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Autumn 1404 (2025)

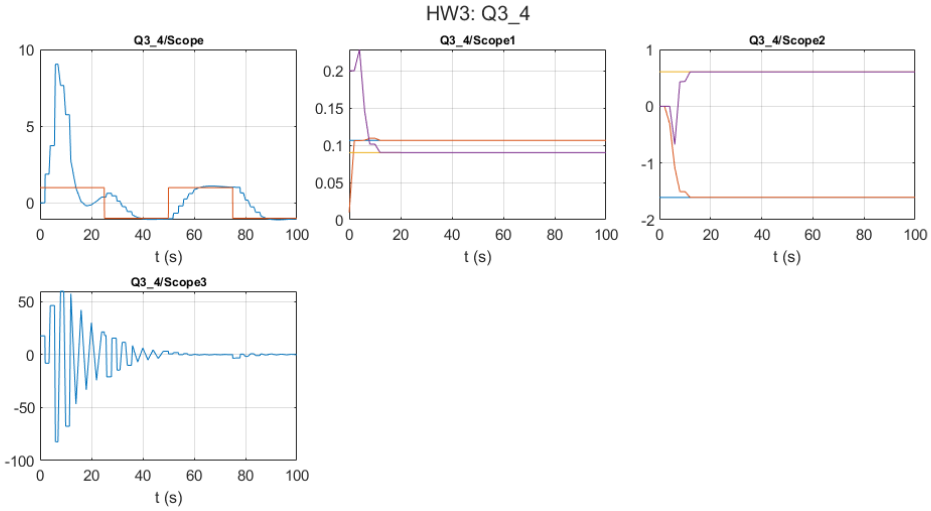
English summary of my original Persian report, which is kept as a PDF in this folder.

All three cases are subsystems of `HW3.slx`. Each one combines RLS estimation of the plant with a pole-placement design recomputed at every step from the current estimates. For each case my report puts my simulation next to the book's figure: output tracking of a square-wave reference, the control signal, and the parameter estimates.

Example 3.4: with zero cancellation. The controller cancels the process zero.

Example 3.5: without zero cancellation. The process zero is kept in the closed-loop response.

Example 3.6: continuous-time STR. The same design applied to a continuous-time plant model.



Re-run in R2023b: example 3.4, output and reference, estimates and control signal.